Electric Vehicle Population

Objective

I analyse a real-world Electric Vehicle Population dataset. This dataset provides detailed information on Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs) currently registered in the United States, specifically through the Washington State Department of Licensing. The dataset includes various attributes of these electric vehicles such as make, model, year of registration and other relevant data that reflects the growing trend of electric vehicle adoption in Washington State.

Approach

- Collected the Electric Vehicle Population dataset containing details such as make, model, model year, electric range, CAFV eligibility and location.
- Replaced or removed null entries in columns like Model Year, Electric Range, and Base MSRP.
- formatted categorical values like CAFV Eligibility using a custom function to classify entries as Eligible, Not Eligible and Unknown.
- Used visualizations histograms, bar charts, Pie chart and count plots.

Outliers and exclusions:

- Missing values for numerical columns were filled with median and Mean values and for object data type columns were filled with Unknown.
- There is 90% of 0 values in column Base MSRP and Electric Range where replace 0 value to Mean Values.

Modelling:

- I use Random Forest Classifier and Logistic Regression Predicting the type of Electric Vehicle Type BEV and PHEV.
- Selected relevant features for modelling such as Model Year, Make, Electric Range, Base MSRP, Legislative District, CAFV Eligibility, State and County. After that Divided the dataset into training and testing sets to evaluate model performance.
- After training the model on the entire dataset, Model predication gives the Random Forest Classifier accuracy is 1.00 and Logistic Regression accuracy is 0.95.

Performance:

Classification Report of Random Forest Classifier:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	9612
1	1.00	1.00	1.00	37527
accuracy			1.00	47139
macro avg	1.00	1.00	1.00	47139
weighted avg	1.00	1.00	1.00	47139

Classification Report of Logistic Regression:

	precision	recall	f1-score	support
0	1.00	0.77	0.87	9612
1	0.94	1.00	0.97	37527
accuracy			0.95	47139
macro avg	0.97	0.88	0.92	47139
weighted avg	0.95	0.95	0.95	47139

Challenges:

- The dataset contained missing values in columns such as Electric Range, Base MSRP, Legislative District.
- There are some columns like Legislative District, Postal Code and Electric Range are in float data type.

Conclusion:

- The project concluded that Battery Electric Vehicles (BEVs) are more likely to be CAFV eligible due to higher electric ranges and recent model years.
- Random Forest achieved 1.00 accuracy, providing reliable insights into EV adoption trends and supporting data-driven environmental planning.
- Battery Electric Vehicles (BEVs) are more likely to qualify for CAFV programs compared to Plug-in Hybrid Electric Vehicles (PHEVs).

Future Improvements:

applying deep learning models or time-series forecasting to predict future EV adoption trends.